CHAPTER 4 · CONCEPT 1 · ANSWER KEY

Angles in Quadrilaterals — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1	**(c) 360°.** Two triangles × 180°.
Q2	**(c) $n(n-3)/2$.** Standard diagonals formula.
Q3	**110°.** 360° − (100° + 80° + 70°) = 110°.
Q4	**(b) 8.** 360° ÷ 45° = 8.
Q5	**True.** Exterior angles of any polygon sum to 360°.
Q6	**(a) Both true, R explains A.** Two triangles give 2 × 180° = 360°.

Short answer

Q7	**36°, 72°, 108°, 144°.** 10 parts = 360° ⇒ 1 part = 36°.
Q8	**12 sides.** 360° ÷ 30° = 12.
Q9	**x = 70°.** 360° − (120° + 80° + 90°) = 70°.

Long answer

Q10	**x = 60°; angles 60°, 120°, 80°, 100°.** $x + 2x + 80 + 100 = 360 \Rightarrow 3x = 180$.
Q11	**(a) 40° (b) 140° (c) 1260°.** Ext = 360⁄9; int = 180 − 40; sum = (9 − 2) × 180.
Q12	**20 diagonals.** $n(n-3)/2 = 8 \times 5/2 = 20$.

Total: 21 marks.